

Deepfake audio detection with spectral features and ResNeXt-based architecture

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ABSTRACT

The increasing prevalence of deepfake audio technologies and their potential for malicious use in fields such as politics and media has raised significant concerns regarding the ability to distinguish fake from authentic audio recordings. This study proposes a robust technique for detecting synthetic audio by leveraging three spectral features: Linear Frequency Cepstral Coefficients (LFCC), Mel Frequency Cepstral Coefficients (MFCC), and Constant Q Cepstral Coefficients (CQCC). These features are processed using an enhanced ResNeXt architecture to improve classification accuracy between genuine and spoofed audio. Additionally, a Multi-Layer Perceptron (MLP)-based fusion technique is employed to further boost the model's performance. Extensive experiments were conducted using three datasets: the ASVspoof 2019 Logical Access (LA) dataset—featuring text-to-speech (TTS) and voice conversion attacks—the ASVspoof 2019 Physical Access (PA) dataset—including replay attacks—and the ASVspoof 2021 LA, PA and DF datasets. The proposed approach has demonstrated superior performance compared to state-of-the-art methods across all three datasets, particularly in detecting fake audio generated by text-to-speech (TTS) attacks. Its overall performance is summarized as follows: the system achieved an Equal Error Rate (EER) of 1.05% and a minimum tandem Detection Cost Function (min-tDCF) of 0.028 on the ASVspoof 2019 Logical Access (LA) dataset, and an EER of 1.14% and min-tDCF of 0.03 on the ASVspoof 2019 Physical Access (PA) dataset, demonstrating its robustness in detecting various types of audio spoofing attacks. Finally, on the ASVspoof 2021 LA dataset the method achieved an EER of 7.44% and min-tDCF of 0.35.

1. Introduction

Recently, the rapid evolution of deep learning techniques has given rise to new and complex challenges in the manipulation of digital media. Image, audio, and video files are increasingly being manipulated using artificial intelligence-based methods known as “deepfake”. Among these, the use of deepfake techniques in audio files is particularly concerning, as it enables the production and dissemination of fake audio recordings—raising serious questions about reliability and security of audio content.

Deepfake audio can convincingly mimic human voices, posing significant threats to the integrity of digital content and exacerbating the challenges of misinformation and disinformation. Most notably, the vulnerability of Automatic Speaker Verification (ASV) systems to such attacks may lead to critical security risks. Consequently, the detection

of deepfake audio has become increasingly important from a cyber-security standpoint. This includes fake speeches, potentially deceptive phone calls, and manipulated digital audio files.

This concern highlights the urgent need for developing effective and reliable methods for detecting deepfake audio, as a means of fostering resilience among individuals, organizations, and societies in general. There are several approaches to creating deepfake audio, with the three most commonly used techniques being text-to-speech (TTS), voice conversion (VC), and replay attacks.

The text-to-speech technique generates synthetic speech from text and clones the target user's voice using a deep learning-based approach, as illustrated in Fig. 1(a). This method is also referred to as speech synthesis (SS). The second technique, voice conversion, involves modifying a speaker's voice to sound like another person using deep learning

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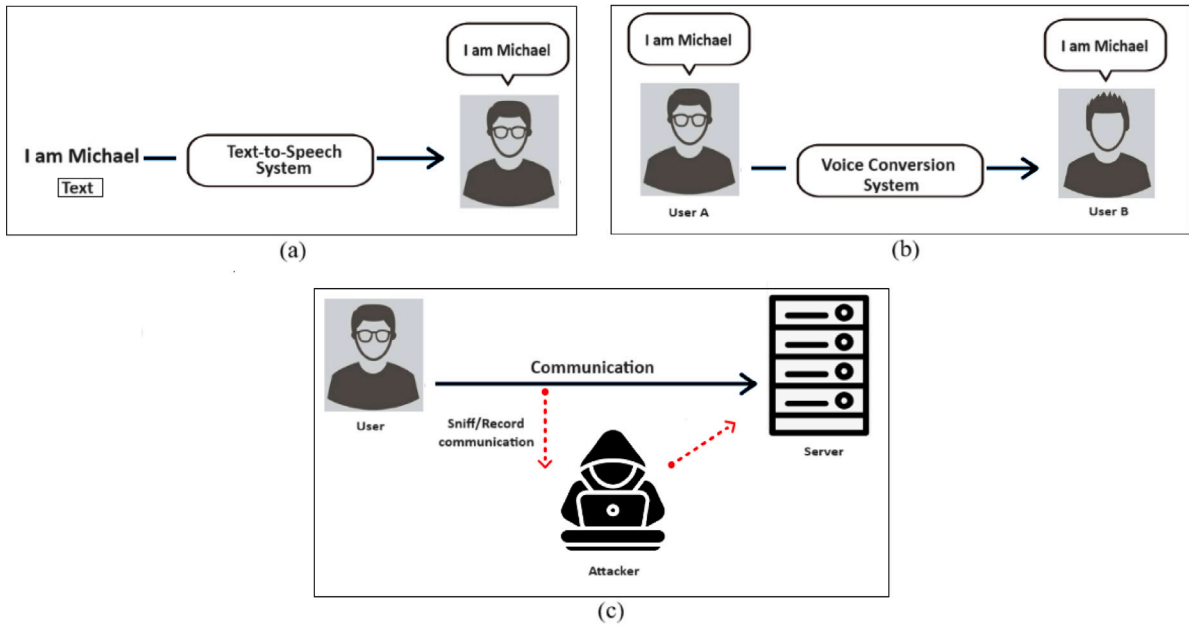


Fig. 1. The most common deepfake audio attack types: (a) TTS, (b) VC, and (c) Replay.

techniques, while preserving the original speech content, as depicted in Fig. 1(b). In a replay attack scenario, shown in Fig. 1(c), an attacker uses a recording device to capture genuine speech and later replays it in an attempt to gain unauthorized access to a system [1].

The issue of deep fake audio posing a threat to Automatic Speaker Verification (ASV) systems was first highlighted by researchers in 2013 [2]. Subsequently, the ASVspoof Challenges have provided a public platform for evaluating detection methods and benchmarking their effectiveness using predefined metrics and standardized datasets.

The first ASVspoof Challenge was organized in 2015, focusing on text-to-speech (TTS), also referred to as speech synthesis (SS), and voice conversion (VC) attacks [3]. In this challenge, the ASVspoof 2015 database was released, and it was suggested that the Equal Error Rate (EER) metric be used for evaluating the performance of proposed methods.

During the ASVspoof 2017 workshop, researchers specifically emphasized that audio replay spoofing attacks are particularly difficult to detect compared to other attack types [1]. The released ASVspoof 2017 dataset includes both original and spoofed audio samples, generated exclusively for replay attack scenarios.

The subsequent challenge, ASVspoof 2019, introduced a new dataset along with a novel evaluation metric called the minimum tandem Detection Cost Function (t-DCF) [4]. The ASVspoof 2019 database consists of two subsets: Logical Access (LA) and Physical Access (PA). While the LA subset contains spoofed audio generated using TTS and VC techniques, the PA subset comprises replayed speech samples.

In the ASVspoof 2021 challenge, the published test dataset expanded the scope of detection tasks beyond LA and PA to include deepfake (DF) detection. All DF audio samples were generated using more than one hundred TTS or VC algorithms [5].

Following this series of challenges, various approaches have been proposed to detect fake audio generated using these techniques. However, the literature consistently highlights the limited generalization ability of current spoofing detection methods. For example, systems that perform well against TTS or VC attacks often struggle with replay attacks [6,7]. In many studies, replay attacks have either been insufficiently addressed or have shown significantly poor performance [8,9].

In this study, we propose a robust technique for detecting synthetic audio. The main contributions of this work are outlined below:

- We demonstrate that Constant Q Cepstral Coefficients (CQCC) perform effectively against TTS and VC attacks, Mel Frequency Cepstral Coefficients (MFCC) are more suitable for replay attacks, and Linear Frequency Cepstral Coefficients (LFCC) provide consistent performance across all attack types. We propose combining these spectral features to improve robustness against a wide range of spoofing scenarios.
- We introduce a novel ResNeXt-based architecture for feature embedding extraction, enhanced with a Multi-Layer Perceptron (MLP)-based classifier for final classification.
- Our proposed approach outperforms state-of-the-art methods on the ASVspoof 2019 LA, ASVspoof 2019 PA, and ASVspoof 2021 LA, PA and DF datasets.

This paper is organized as follows: Section 2 provides a review of existing approaches and methodologies for detecting deep fake audio, with a focus on categorizing them based on their feature extraction techniques. Section 3 details the proposed detection method. Section 4 describes the experimental setup and presents a comparative performance analysis between the proposed method and state-of-the-art techniques. Finally, Section 6 concludes the paper and discusses potential directions for future research.

2. Related works

Deep fake audio detection models typically comprise two main components: front-end feature extraction and back-end classification. However, recent studies have also introduced models employing end-to-end architectures. This section provides a review of the methods in the existing literature, categorizing them into three primary groups: feature engineering approaches, classification models, and end-to-end architectures.

2.1. Feature engineering

The first step in a deepfake audio detection method is the extraction of features from the input audio. Feature extraction approaches can generally be classified into two categories: hand-crafted spectral features and deep learning-based features. Among the hand-crafted

features, the most commonly used are magnitude-based spectral coefficients, phase-based spectral coefficients, and image-like representations.

The majority of front-end features in the literature are primarily based on short-term spectral coefficients such as Mel-Frequency Cepstral Coefficients (MFCC) [10–15], Inverted Mel-Frequency Cepstral Coefficients (IMFCC) [12,16,17], Linear Frequency Cepstral Coefficients (LFCC) [18,19], Short-Time Fourier Transform (STFT) [15,18,20], and Gammatone Cepstral Coefficients (GTCC) [11].

Alzantot et al. (2019) [15] conducted a comparative study on Log-STFT, MFCC, and Constant Q Cepstral Coefficients (CQCC) using a ResNet-based architecture, reporting performance results for each feature set individually.

In contrast to short-term spectral features that rely on fixed and narrow window lengths, long-term window transforms have been proposed to capture long-range temporal dependencies and provide higher frequency resolution. One such feature is the CQCC, which has demonstrated strong performance in multiple studies [14,21].

To investigate the impact of phase information in spoofing detection, phase-based features have also been employed, such as Modified Group Delay Cepstral Coefficients (MGDCC) [22] and relative phase features. In other studies, spectrogram-based representations – providing detailed insights into the frequency and intensity variations of the audio signal over time – have been utilized. Specifically, Mel-spectrograms (Mel-Spec) [23] and Constant-Q Transform spectrograms (CQT-Spec) [24] have been exploited as input features for deep learning models.

With the evolution of deep learning techniques, several methods have emerged that utilize learnable features to capture the underlying characteristics of audio signals. One of the earliest studies in this category extracted two types of deep features using Deep Neural Network-based frame-level and Recurrent Neural Network (RNN)-based sequence-level feature extraction approaches [25].

Lavrentyeva et al. [26] proposed a method that combines spectral features with deep feature extraction using i-vectors and Light Convolutional Neural Networks (LCNN). Similarly, Li et al. [27] introduced an enhanced Res2Net architecture by integrating a channel-wise gating mechanism into the connections among feature groups, resulting in the Channel-Wise Gated Res2Net (CG-Res2Net).

Wu et al. [28] developed an LCNN model incorporating a feature genuinization strategy, while Wang et al. [29] investigated the use of MFCC, spectral, and Constant-Q Transform (CQT) features with a DenseNet-based architecture. Li et al. (2020) further extended this line of work by proposing multiple variants of the Channel-wise Gate-Res2Net50 model—namely, Single-group Channel-wise Gate (SCG-Res2Net50), Multi-group Channel-wise Gate (MCG-Res2Net50), and Multi-group Latent-space Channel-wise Gate (MLCG-Res2Net50)—using CQT features [27].

Xue et al. [30] explored a multi-modal approach by fusing LFCC and facial features using the SE-DenseNet model [31], combined with A-softmax for classification. Wang et al. [32] studied the fusion of Multi-scale Permutation Entropy (MPE) and LFCC features with SENet and LCNN architectures. Additionally, Chaiwongyen et al. [33] proposed fusing pathological features (PF) with Mel-spectrogram inputs using a ResNet18 model.

Another line of research involves U-Net architectures enhanced with attention mechanisms and skip connections [34]. Furthermore, methods based on pre-trained self-supervised learning (SSL) models – such as wav2vec 2.0 [35,36], WavLM [37,38], HuBERT [39], and the Whisper encoder [40] – are also categorized within this group.

2.2. Classification models

Both traditional machine learning and deep learning models – such as Convolutional Neural Networks (CNN) and Residual Networks

(ResNet) – have been employed for the classification of extracted features in deepfake audio detection. Among machine learning-based classifiers, Support Vector Machines (SVM) and Gaussian Mixture Models (GMM) have been commonly used [41–45].

Lavrentyeva et al. [46] utilized a CNN architecture to address the anti-spoofing problem, proposing a lightweight implementation through the use of a Light Convolutional Neural Network (LCNN). ResNet-based models are also widely adopted in the classification phase [47,48], while DenseNet-based architectures have been explored for the same task [49,50].

2.3. End-to-end (E2E) architectures

Methods in this category are integrated into a unified architecture that spans from the input layer to the final output, eliminating the need for separate feature extraction and classification phases.

The method proposed by Ma et al. [51] employs raw audio waveforms as input and applies a 1-D convolutional layer with a kernel size of 3, followed by ResNet blocks. Similarly, studies [52,53] adopt a joint architecture based on convolutional Long Short-Term Memory (LSTM) neural networks, operating directly on raw audio inputs.

Jung et al. [54] proposed another method based on a variant of the RawNet2 encoder. They proposed an end-to-end (E2E) model with a graph attention layer for heterogeneous stacking, capturing different temporal and spectral domains, was proposed. Tak et al. [55] also used RawNet2 with raw audio forms. Hua et al. [56] proposed an E2E ResNet-based approach.

Other methods also leverage pre-trained self-supervised learning models with or without data augmentation [57] used HuBERT (Hidden-Unit BERT) model to extract audio feature. While [58] proposed to use Wav2Vec2, [59] utilized WavLM and [60] opted WavLM-Base, Wav2Vec2-Large, and HuBERT-Base with different classification and fusion strategies.

While end-to-end (E2E) architectures and SSL models have gained popularity due to their ability to learn hierarchical representations directly from raw audio, they exhibit notable limitations in handling replay-based PA attacks, which are frequently encountered in real-world scenarios. The lack of explicit spectral priors and the overfitting tendency on training domains often reduce their robustness against unseen spoofing conditions.

Therefore, rather than adopting an E2E architecture, the proposed approach emphasizes spectral feature extraction as an explicit front-end process. These features are processed through a ResNeXt-based model to learn high-level embeddings across multiple datasets. Subsequently, a feature embedding fusion strategy is employed to enhance generalization by integrating complementary information from different spectral cues. This hybrid design aims to balance interpretability, robustness, and transferability under varied PA conditions.

3. Proposed method

This research aims to develop a technique capable of determining whether an audio recording is genuine or synthetically generated.

In this section, we present our proposed method, which utilizes three distinct spectral features in combination with an improved ResNeXt-based model. The model is independently trained three times, once for each feature type, to generate individual classification scores. These scores are subsequently fused using a Multi-Layer Perceptron (MLP)-based approach to produce the final classification result.

The overall architecture of the proposed method is illustrated in Fig. 2. It consists of three main components: (1) the spectral feature extraction module, (2) the improved ResNeXt-based classification module, and (3) the MLP-based score fusion module. The details of each component are provided in the following subsections.

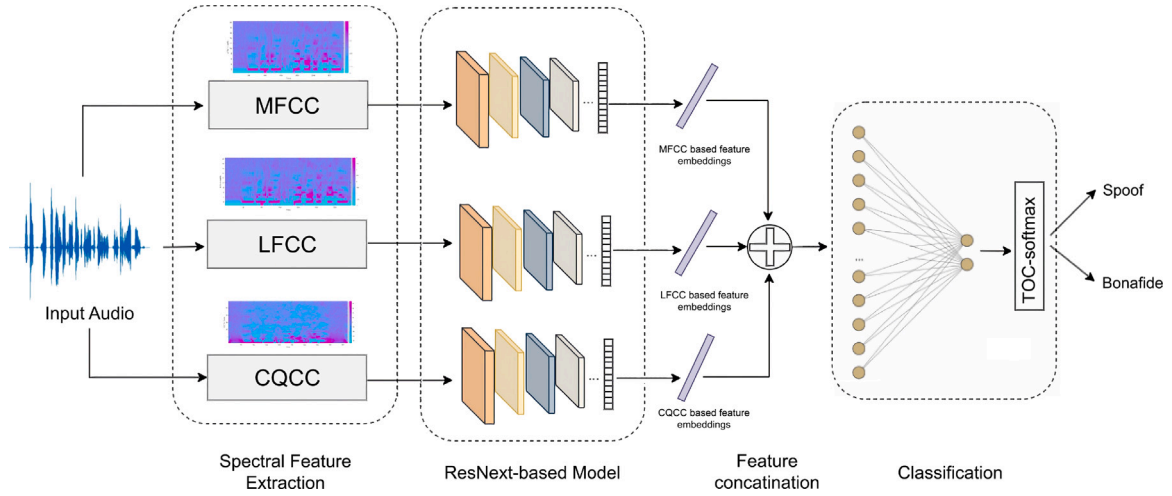


Fig. 2. The overview of the proposed system.

3.1. Spectral feature extraction module

The input audio is represented as a feature matrix composed of three spectral feature types: Mel-Frequency Cepstral Coefficients (MFCCs), Linear Frequency Cepstral Coefficients (LFCCs), and Constant Q Cepstral Coefficients (CQCCs). These coefficients have proven effective in various audio processing applications due to their ability to extract robust and discriminative characteristics across multiple frequency bands.

Although the general extraction procedures for MFCC and LFCC are similar, they differ in their frequency scaling strategies. MFCC applies a nonlinear Mel scale after computing the power spectrum from the Fourier transform, whereas LFCC uses a linear frequency scale following the computation of the magnitude spectrum. The Mel scale emphasizes lower frequencies, aligning with the higher sensitivity of human hearing in this range, making MFCC particularly effective in speech and speaker recognition tasks. In contrast, the linear frequency scale in LFCC provides uniform spectral resolution across all frequency bands, which is advantageous for applications where capturing fine-grained spectral details across the full frequency range is critical.

Unlike MFCCs and LFCCs, Constant Q Cepstral Coefficients (CQCCs) are derived from the Constant Q Transform (CQT), which provides a variable-frequency-resolution analysis—offering higher resolution at lower frequencies and lower resolution at higher frequencies. This property makes CQCCs particularly suitable for capturing detailed spectral patterns in the lower frequency range.

To leverage the unique advantages of each feature type, it is proposed that MFCCs, LFCCs, and CQCCs be used independently during model training. This strategy allows the model to benefit from the complementary characteristics offered by different frequency scaling techniques.

The following subsections provide a comprehensive overview of the extraction methods for each of these three spectral coefficient types.

3.1.1. MFCCs extraction

Mel-Frequency Cepstral Coefficients (MFCCs), originally proposed by Davis and Mermelstein in 1980 [61], are a widely used feature representation technique in speech and audio signal processing.

The input audio signal $x[n]$ is first pre-emphasized to amplify its high-frequency components. This step serves to balance the frequency spectrum and enhance the signal-to-noise ratio, which is beneficial for the subsequent analysis stages. Pre-emphasis is typically performed using a first-order high-pass filter defined as:

$$y[n] = x[n] - \alpha x[n-1] \quad (1)$$

where $x[n]$ is the original input signal, and α is the pre-emphasis coefficient, commonly set in the range $0.95 \leq \alpha \leq 0.97$.

The computation of MFCCs involves several sequential steps, as outlined below.

Step 1. Framing and Windowing: The pre-emphasized signal $x[n]$ is segmented into overlapping frames of N samples, with a frame shift of M samples. The frame length is computed as $N = T \times f_s$, where f_s denotes the sampling rate and T is the frame duration. In this study, $T = 20$ ms and the frame shift is set to 10 ms.

Each frame $y[n]$ is then multiplied by a window function $w[n]$ to reduce spectral leakage, commonly using a Hamming window. In this study, the window length is set to 25 ms. The windowed frame is computed as:

$$x_m[n] = y[n] \cdot w[n], \quad 0 \leq n < N \quad (2)$$

The Hamming window is defined in Eq. (3) [62]:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) \quad (3)$$

Step 2. Fast Fourier Transform (FFT): Each windowed frame $x_m[n]$ is transformed into its frequency-domain representation $X_m[k]$ using the Fast Fourier Transform (FFT), defined as:

$$X_m[k] = \sum_{n=0}^{N-1} x_m[n] \cdot e^{-j \frac{2\pi kn}{N}}, \quad 0 \leq k < N \quad (4)$$

The corresponding magnitude spectrum is computed by taking the absolute value of the FFT output:

$$|X_m[k]| = \left| \sum_{n=0}^{N-1} x_m[n] \cdot e^{-j \frac{2\pi kn}{N}} \right| \quad (5)$$

$$|X_m[k]| = \sqrt{(\text{Re}\{X_m[k]\})^2 + (\text{Im}\{X_m[k]\})^2} \quad (6)$$

Here, j denotes the imaginary unit, and N represents the FFT size [63]. In this study, $N = 256$ is used.

Step 3. Mel-Scale Filter Bank Processing: Next, the magnitude spectrum is passed through a Mel-scale filter bank consisting of $K = 13$ triangular filters, denoted as $H_k(m)$, which are spaced non-linearly according to the Mel frequency scale.

The Mel scale is designed to mimic human auditory perception by allocating more filters to lower frequencies where the human ear is more sensitive. It is defined by the following equation [64]:

$$f_{\text{Mel}} = 2595 \log_{10} \left(1 + \frac{f}{700} \right) \quad (7)$$

The filter bank is designed to span the frequency range $[0, f_{\text{max}}]$ Hz, where f_{max} is set to half of the sampling rate, also known as the Nyquist frequency.

The center frequencies m_k of the Mel filters are uniformly spaced in the Mel frequency domain and subsequently converted back to the linear frequency domain for implementation. In this study, the number of Mel filters is set to 26 to ensure adequate frequency resolution across the perceptual spectrum.

The center frequency m_k (in terms of FFT bin index) corresponding to each Mel-scaled frequency is computed as:

$$m_k = \frac{f_{\text{Mel}} \cdot N}{f_s} \quad (8)$$

where f_{Mel} is the center frequency on the Mel scale, N is the FFT size, and f_s is the sampling rate.

Each triangular filter $H_k(m)$ in the Mel filter bank is then defined as:

$$H_k(m) = \begin{cases} \frac{m-m_{k-1}}{\Delta m_k} & \text{if } m_{k-1} \leq m \leq m_k \\ \frac{m_{k+1}-m}{\Delta m_k} & \text{if } m_k < m \leq m_{k+1} \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where $\Delta m_k = m_{k+1} - m_{k-1}$ denotes the width of the k th triangular filter in the frequency domain. After applying each filter $H_k(m)$ to the power spectrum $|X_m[k]|^2$, the filter bank energies E_k are computed as:

$$E_k = \sum_{m=0}^{N-1} |X_m[k]|^2 \cdot H_k(m) \quad (10)$$

Step 4. Logarithmic Compression: To mimic the human auditory system's nonlinear perception of loudness and to reduce dynamic range variations, the filter bank energies E_k are subjected to logarithmic compression:

$$L_k = \log(E_k + \epsilon) \quad (11)$$

where ϵ is a small constant added to avoid numerical instability when E_k approaches zero.

Step 5. Discrete Cosine Transform (DCT): The final step applies the Discrete Cosine Transform (DCT) to the log-compressed filter bank energies in order to decorrelate the features and obtain the Mel-Frequency Cepstral Coefficients (MFCCs). The DCT is computed as:

$$\text{MFCC}_n = \sum_{m=1}^K \log(E_m) \cdot \cos \left[\frac{\pi n}{K} \left(m - \frac{1}{2} \right) \right] \quad (12)$$

Here, $1 \leq n \leq D$, where MFCC_n denotes the n th cepstral coefficient, and D is the number of coefficients retained. In this study, we set $D = 13$.

In addition to static features, audio signals also contain dynamic characteristics, which are captured by computing the first- and second-order temporal derivatives of the MFCCs, denoted as ΔMFCC_n (delta MFCC) and $\Delta \Delta \text{MFCC}_n$ (delta-delta MFCC), respectively. These dynamic features model short-term temporal variations in the audio signal and provide complementary information to the static MFCC coefficients.

Incorporating these derivatives enhances the robustness and discriminative power of the feature representation, particularly in tasks such as speech and speaker recognition.

The final MFCC-based feature matrix is defined as:

$$\text{FM}_{\text{MFCC}} = [\text{MFCC}_n, \Delta \text{MFCC}_n, \Delta \Delta \text{MFCC}_n] \quad (13)$$

3.1.2. LFCCs extraction

In the proposed method, Linear Frequency Cepstral Coefficients (LFCCs) are used as an additional spectral feature for audio representation. As previously discussed, LFCCs are designed to capture the spectral characteristics of audio signals. While MFCCs utilize a Mel-scaled filter bank to emphasize perceptually important low-frequency regions, LFCCs employ a linearly spaced filter bank, making them more suitable for applications requiring uniform resolution across the frequency spectrum.

In this study, the LFCC extraction process begins with pre-emphasis using a coefficient $\alpha = 0.97$. The initial steps mirror those in the MFCC

extraction process, with a frame length of $T = 20$ ms. However, for the FFT stage, a larger FFT size of $N = 512$ is used, following the ASvspoof 2019 baseline configuration [4].

In the filter bank construction stage, instead of the Mel scale, a linearly spaced filter bank is applied, based on the sampling frequency f_s . The corresponding linear frequency scale is defined as [63]:

$$f_{\text{linear}}(k) = \frac{k}{N} f_s, \quad \text{for } k = 0, 1, 2, \dots, N-1 \quad (14)$$

Here, N , the number of frequency bins in the filter bank, is set to $N = 20$. Unlike MFCCs, which transform the linear frequency scale to the Mel scale before applying the filter bank, the LFCC approach applies the filter bank directly on the linear frequency scale without any transformation.

This distinction results in MFCCs being more sensitive to perceptually relevant low-frequency bands, whereas LFCCs provide a more uniform spectral resolution across the entire frequency range, particularly in the low-frequency region.

Finally, the complete LFCC-based feature matrix, including the static, delta, and delta-delta coefficients, is represented as:

$$\text{FM}_{\text{LFCC}} = [\text{LFCC}_n, \Delta \text{LFCC}_n, \Delta \Delta \text{LFCC}_n] \quad (15)$$

3.1.3. CQCCs extraction

As an alternative to traditional Fourier-based time-frequency analysis, the Constant Q Cepstral Coefficients (CQCCs) have been proposed—particularly for applications such as deepfake audio detection [2]—to effectively analyze signals with non-uniform spectral characteristics. This method integrates the Constant Q Transform (CQT) with cepstral analysis to generate a more perceptually meaningful feature representation.

Compared to Mel-Frequency Cepstral Coefficients (MFCCs), CQCCs provide a time-frequency resolution that better aligns with human auditory perception, making them a promising alternative. The CQT employed in CQCC computation uses geometrically spaced center frequencies while preserving a fixed center frequency-to-bandwidth ratio. This structure enables higher frequency resolution in lower frequencies and enhanced temporal resolution in higher frequencies, outperforming traditional Fourier-based approaches in these aspects.

Due to its logarithmic frequency scale, the CQT offers a more accurate spectral representation of signals that include both low- and high-frequency components, making it particularly useful in complex audio analysis tasks. Implementing CQCCs in speaker and deepfake detection systems can improve overall performance by extracting richer and more informative spectral features from the audio signal.

Step 1. Constant-Q Transform (CQT): The Constant-Q Transform of a discrete-time signal is defined as:

$$X_{\text{CQT}}(k, m) = \sum_{n=0}^{N-1} x(n) \cdot w(n, k) \cdot e^{-j2\pi f_k n} \quad (16)$$

Here, $X_{\text{CQT}}(k, m)$ denotes the CQT coefficient at frequency bin k and time frame m ; $x(n)$ is the input signal; $w(n, k)$ is a time- and frequency-dependent window function; and f_k represents the logarithmically spaced center frequencies of the CQT filter bank.

In this transformation, the frequency resolution increases as frequency decreases, offering finer detail at lower frequencies and coarser resolution at higher frequencies. This property enables the CQT to provide consistent coverage across a logarithmic frequency scale—an important characteristic for capturing perceptually relevant information in audio signals such as speech and music. As a result, CQT is particularly well-suited for applications involving human auditory perception.

Step 2. Logarithmic Compression: The next step involves applying logarithmic compression to the power spectrum of the Constant-Q Transform (CQT) coefficients. This non-linear transformation reduces the dynamic range of the signal and mimics the human auditory system's logarithmic perception of loudness.

The compressed CQT coefficients are computed as:

$$X'_{\text{CQT}}(k, m) = \log(|X_{\text{CQT}}(k, m)|^2 + \epsilon) \quad (17)$$

where ϵ is a small constant added to prevent numerical instability when the magnitude approaches zero.

where $X'_{\text{CQT}}(k, m)$ is the log-compressed magnitude of the Constant-Q Transform coefficients, and ϵ is a small constant used to avoid taking the logarithm of zero.

Step 3. Uniform Resampling: Since the CQT frequency bins are logarithmically spaced and not uniformly distributed, the log-compressed coefficients $X'_{\text{CQT}}(k, m)$ are interpolated onto a uniform frequency grid. This resampling enables consistent spacing for the subsequent Discrete Cosine Transform (DCT) step. Spline interpolation is commonly used for this purpose:

$$X'_{\text{CQT}}(l, m)_{\text{interp}} = \text{Interpolate} \left\{ X'_{\text{CQT}}(k, m) \right\} \quad (18)$$

Here, $l = 0, 1, \dots, L-1$, and L denotes the number of uniformly spaced frequency bins after interpolation.

Step 4. Discrete Cosine Transform (DCT): Finally, the Discrete Cosine Transform is applied to the interpolated log-compressed CQT features to decorrelate them and obtain the Constant Q Cepstral Coefficients (CQCCs). The DCT is defined as:

$$\text{CQCC}_n = \sum_{k=0}^{K-1} X'_{\text{CQT}}(l, m)_{\text{interp}} \cdot \cos\left(\frac{\pi n(2k+1)}{2K}\right) \quad (19)$$

where $n = 0, 1, \dots, D-1$, and D is the number of cepstral coefficients retained.

where c_n denotes the resulting Constant-Q Cepstral Coefficients (CQCCs), and K is the number of CQT bins per octave. In this study, we set $K = 96$, following the configuration in the ASVspoof baseline system.

The final CQCC-based feature representation is a 90-dimensional matrix, which includes static, delta, and delta-delta coefficients to capture both spectral and temporal dynamics. It is defined as:

$$\text{FM}_{\text{CQCC}} = [\text{CQCC}_n, \Delta\text{CQCC}_n, \Delta\Delta\text{CQCC}_n] \quad (20)$$

3.2. Improved ResNeXt-based model

The ResNeXt architecture is an enhanced version of the ResNet model, originally proposed for image classification tasks [65]. Beyond image classification, ResNeXt has been successfully applied in various domains, particularly in deep learning tasks involving both visual and textual data.

Zhou et al. [66] investigated the effectiveness of the ResNeXt architecture for speaker verification tasks, demonstrating its strong performance on audio-based problems. These findings indicate that ResNeXt is a promising model for applications involving speech and audio signal processing.

A ResNeXt block is built upon the split-transform-merge strategy, where a set of parallel transformations is applied to partitions of the input feature map. The architecture consists of repeated modular blocks that decompose high-dimensional convolutions into several lower-dimensional grouped convolutions, each sharing the same topological structure.

The outputs of these grouped convolutions are then aggregated – typically through summation or concatenation – to form the final feature representation. The concept of cardinality refers to the number of groups into which the original high-dimensional convolution is split. Increasing the cardinality has been shown to improve model performance more effectively than simply increasing the depth or width of the network, both of which also contribute to performance but often at a higher computational cost.

This design enables the model to capture more diverse and complex feature patterns while maintaining computational efficiency [65].

The ResNeXt block can also be interpreted in terms of grouped convolutions. Specifically, multiple parallel 1×1 convolutional layers can be replaced by a single, wider 1×1 grouped convolutional layer. Grouped convolution divides the input channels into several branches (groups), each of which processes a subset of the input independently. This structure naturally implements the split-transform-merge paradigm required by ResNeXt.

In this context, the values typically specified in parentheses represent the number of input channels, the size of the convolution kernel, and the number of convolution kernels, respectively. The parameter c denotes the number of groups in the grouped convolution, which corresponds to the cardinality of the ResNeXt block. A higher cardinality enables the network to learn more diverse representations while maintaining computational efficiency.

The structure of the classification network is illustrated in Fig. 3 (right). The model begins with a 2D convolutional layer with a filter size of 3×3 , 64 convolutional kernels, and a stride of 1. This layer is followed by Batch Normalization (BN) and a ReLU activation function to introduce non-linearity. A max-pooling operation is applied to reduce the spatial resolution.

Subsequently, the network employs a sequence of ResNeXt blocks (also depicted in Fig. 3 (right)), each designed to extract increasingly abstract and complex features. The number of convolutional kernels in the successive ResNeXt stages is 64, 128, 256, and 512, respectively.

Each ResNeXt block adopts a bottleneck architecture, which includes: - a 1×1 convolution for dimensionality reduction, - a grouped 3×3 convolution for parallel processing within groups (with cardinality c), - and another 1×1 convolution to restore the original channel dimension.

Batch Normalization (BN) is applied after each convolutional layer to stabilize learning, and a residual (skip) connection adds the block's input to its output, followed by a ReLU activation function. This configuration enables the network to learn deeper representations with fewer parameters compared to standard ResNet architectures.

After the final ResNeXt block, the model uses an adaptive average pooling layer, which automatically adjusts the output dimensions based on the input size. A flattening layer is then applied to generate the final feature embeddings. The complete layer-wise configuration of the network is presented in Table 1.

3.3. Classification

Up to this stage, three distinct spectral features, MFCC, LFCC, and CQCC, were extracted from the input audio signal. Each of these feature sets was independently processed through the ResNeXt-based architecture to generate high-level feature embeddings.

The resulting embeddings were then concatenated, as shown in Eq. (21), to form a comprehensive and unified feature vector. This concatenation strategy effectively integrates the complementary information captured by each individual spectral representation, leading to a richer and more discriminative feature space for classification.

$$FE_i = \begin{bmatrix} FE_{(i,1)} \\ FE_{(i,2)} \\ FE_{(i,3)} \end{bmatrix} \quad (21)$$

In the proposed study, the classification of the extracted feature embeddings FE_i is performed to improve the overall performance of the detection system, with a specific focus on anti-spoofing measures in speaker recognition.

To this end, a Multilayer Perceptron (MLP)-based classification model is employed. The model comprises a single fully connected hidden layer with 100 neurons, followed by a Rectified Linear Unit (ReLU) activation function, and a fully connected output layer with 64 neurons.

For the training objective, an enhanced version of the OC-Softmax loss—referred to as the Thresholding One-Class Softmax (TOC-Softmax) loss—is utilized [67]. This loss function is specifically designed to mitigate the inconsistency introduced by class imbalance, which is a common issue in datasets such as ASVspoof 2019.

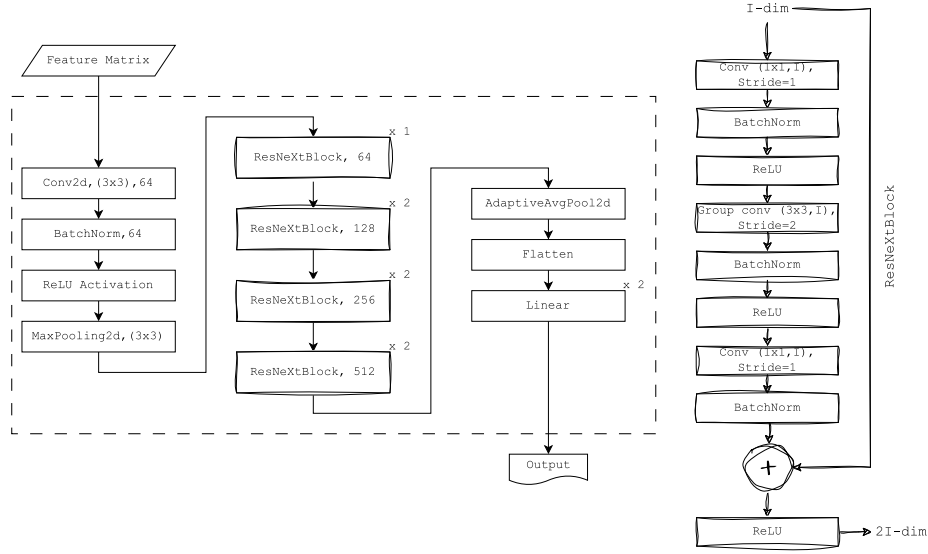


Fig. 3. Left: Overall structure of the proposed architecture. Right: The used ResNextBlock.

Table 1

Layer-wise details of the model configuration.

Layer	Filter size/Filters/Stride	Output
Input	–	(1, 60, 750)
Conv0	Conv2d (3 × 3, 64, 1)	(64, 60, 750)
BN0	BN (64)	(64, 60, 750)
Pool0	MaxPool2d (3 × 3, 2)	(64, 30, 375)
ResNeXtBlock1	$\left\{ \begin{array}{l} \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN \& ReLU} \\ \text{GroupConv2d}(3 \times 3, 64, 2) \\ \text{BN \& ReLU} \\ \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN} \end{array} \right\} \times 1$	(128, 30, 375)
ResNeXtBlock2	$\left\{ \begin{array}{l} \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN \& ReLU} \\ \text{GroupConv2d}(3 \times 3, 64, 2) \\ \text{BN \& ReLU} \\ \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN} \end{array} \right\} \times 2$	(256, 15, 188)
ResNeXtBlock3	$\left\{ \begin{array}{l} \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN \& ReLU} \\ \text{GroupConv2d}(3 \times 3, 64, 2) \\ \text{BN \& ReLU} \\ \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN} \end{array} \right\} \times 2$	(512, 8, 94)
ResNeXtBlock4	$\left\{ \begin{array}{l} \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN \& ReLU} \\ \text{GroupConv2d}(3 \times 3, 64, 2) \\ \text{BN \& ReLU} \\ \text{Conv2d}((1 \times 1), 64, 1) \\ \text{BN} \end{array} \right\} \times 2$	(1024, 4, 47)
AdaptiveAvgPool2d	AdaptiveAvgPool2d (1 × 1, 1)	(1024, 1, 1)
Flatten	–	(1024)

4. Experimental results and discussion

This section provides a comprehensive overview of the experiments conducted to evaluate the effectiveness of the proposed method. Furthermore, a comparative analysis is presented to benchmark the performance of the proposed approach against existing state-of-the-art methods reported in the literature.

Table 2

Number of utterance in ASVspoof 2019 LA, ASVspoof 2019 PA and ASVspoof 2021 LA datasets.

Subsets	ASVspoof 2019 LA		ASVspoof 2019 PA		ASVspoof 2021 LA	
	Spoof	Bonafide	Spoof	Bonafide	Spoof	Bonafide
Train	22,800	2580	48,600	5400	–	–
Dev	22,296	2548	24,300	5400	–	–
Eval	63,882	7355	135,000	8000	137,457	7355

4.1. Dataset

The performance of the proposed method was evaluated on three benchmark datasets: the ASVspoof 2019 Logical Access (LA), Physical Access (PA) [4], and the ASVspoof 2021 LA dataset [5].

The ASVspoof 2019 LA dataset includes a wide range of spoofing attack algorithms designed to simulate various types of audio spoofing techniques. These algorithms fall into three main categories: TTS, voice conversion VC, and hybrid methods.

TTS algorithms synthesize speech directly from text by leveraging deep learning models and neural vocoders, resulting in highly natural and intelligible speech signals. VC (Voice Conversion) algorithms, in contrast, transform existing human speech to resemble the voice characteristics of another speaker by mapping spectral features—typically using statistical models or deep neural networks.

The dataset includes original utterances from both male and female speakers, which serve as source material for generating VC-based synthetic speech.

Hybrid methods in the ASVspoof 2019 LA dataset combine both TTS and VC techniques to further enhance the naturalness and diversity of spoofed audio. These approaches integrate neural architectures and advanced signal processing methods to generate high-fidelity spoofing attacks.

In total, the dataset contains 19 distinct spoofing attack types, labeled A01 through A19. The training and development subsets consist of attacks A01–A06, while the evaluation subset contains attacks A07–A19. A detailed breakdown of the number of utterances in each subset is provided in Table 2.

In the evaluation set of the ASVspoof 2019 LA dataset, spoofing attacks labeled A16 and A19 are categorized as known attacks. This

classification is due to the fact that the same underlying algorithms (A04 and A06) used to generate these attacks were also employed in the training and development subsets, although with different speakers and audio samples.

Attacks labeled A10–A15 and A18 are considered unknown attacks, as their generating algorithms are entirely absent from the training and development sets. Meanwhile, A07–A09 and A17 are categorized as partially known attacks, as they involve variations of known algorithms or closely related spoofing strategies [4].

The ASVspoof 2019 Physical Access (PA) dataset includes replay attacks and was created using a stationary microphone setup in a controlled acoustic environment. The nature of each attack depends on two primary factors: (1) the distance between the loudspeaker (attacker) and the microphone (ASV system), and (2) the quality of the playback and recording devices. These conditions are encoded using two-character labels such as AA, AB, ..., CB, CC [4].

Another dataset used in the evaluation are the ASVspoof 2021 Logical Access (LA), Physical Access (PA) and Deepfake (DF) sets [5]. Building upon the ASVspoof 2019 scenarios, this dataset introduces more complex and diverse spoofing attacks designed to evaluate the generalization capabilities of anti-spoofing systems. ASVspoof 2021 LA, DF and PA contains only an evaluation set; the training and development subsets are inherited from the ASVspoof 2019 LA dataset for ASVspoof 2021 LA and ASVspoof 2021 DF; from the ASVspoof 2019 PA dataset for ASVspoof 2021 PA.

The ASVspoof 2021 LA evaluation set also introduces a transmission variability component, simulating real-world deployment environments. The spoofed and bona fide audio samples are transmitted over telephony networks such as PSTN and VoIP, encoded with various codecs, sampling rates, and bitrates. These conditions are labeled C1 through C7 [5], allowing for a thorough assessment of system robustness under channel and codec mismatches. The spoofed audio in the DF set closely resembles that in the LA set; however, DF utterances underwent audio data compression and decompression using alternative audio codecs rather than telephone codecs. On the other hand, spoofed utterances in the ASVspoof 2021 PA set include both genuine and artificially generated recordings, created by replaying and capturing authentic utterances through various electronic devices across different acoustic environments.

The ultimate objective of these datasets is to encourage the development of anti-spoofing countermeasures that maintain high detection accuracy even in the presence of unseen and challenging attack scenarios.

4.2. Evaluation metrics

To evaluate the performance of the proposed system, we adopt two widely used metrics: Equal Error Rate (EER) and minimum tandem Detection Cost Function (min-tDCF).

EER serves as a general indicator of verification accuracy, representing the point at which the false acceptance rate (FAR) and false rejection rate (FRR) are equal [4]. However, EER alone may not fully capture the performance of systems deployed in real-world applications.

Therefore, min-tDCF is also employed, as it reflects the combined impact of errors from both the Automatic Speaker Verification (ASV) system and the countermeasure (CM) module. This metric provides a more realistic measure of system performance under integrated deployment scenarios. Its use aligns with the official recommendation of the ASVspoof 2019 Challenge, where min-tDCF is designated as the default evaluation metric [68], thereby ensuring that our results remain consistent and comparable with prior benchmarks.

EER is computed through the False Rejection Rate (FRR) and False Acceptance Rate (FAR) at a certain threshold (γ). These rates can be described as follows:

$$FRR(\gamma) = \frac{\#\{\text{Original audios with score} \leq \gamma\}}{\#\{\text{Total original audios}\}} \quad (22)$$

$$FAR(\gamma) = \frac{\#\{\text{Fake audios with score} > \gamma\}}{\#\{\text{Total fake audios}\}} \quad (23)$$

Here, the score refers to the output of the classifier. A higher score indicates a higher likelihood of the input being genuine (original) audio, while a lower score suggests a spoofed (fake) audio signal.

Rather than selecting an arbitrary threshold γ , the threshold can be set to θ_{eer} , which satisfies the following condition:

$$FAR(\theta_{\text{eer}}) = FRR(\theta_{\text{eer}}) \quad (24)$$

This threshold θ_{eer} defines the point at which the system's false acceptance and false rejection rates are equal, yielding the EER.

$$EER = FRR(\theta_{\text{eer}}) = FAR(\theta_{\text{eer}}) \quad (25)$$

Another key metric used for evaluating the system's performance is the minimum normalized tandem Detection Cost Function ($\min_t\text{-DCF}$). This metric provides a comprehensive assessment of the integrated performance of the Automatic Speaker Verification (ASV) system and the countermeasure (CM) module.

Unlike standalone metrics such as EER, $\min_t\text{-DCF}$ accounts for the interaction between ASV and CM errors under predefined cost and prior parameters. As such, it offers a more realistic evaluation of system security in practical deployment scenarios.

The mathematical formulation of the $\min_t\text{-DCF}$ is given in Eq. (26).

$$\min_t\text{-DCF} = \min_{\tau_{\text{cm}}} \left\{ \frac{C_0 + C_1 P_{\text{miss}}^{\text{cm}}(\tau_{\text{cm}}) + C_2 P_{\text{fa}}^{\text{cm}}(\tau_{\text{cm}})}{t\text{-DCF}_{\text{default}}} \right\} \quad (26)$$

Let $P_{\text{miss}}^{\text{cm}}(\tau_{\text{cm}})$ and $P_{\text{fa}}^{\text{cm}}(\tau_{\text{cm}})$ denote the miss rate and false alarm rate, respectively, of the countermeasure (CM) system at a given decision threshold τ_{cm} . These rates reflect the system's performance in rejecting spoofed inputs and accepting genuine inputs.

The cost parameters C_0 , C_1 , and C_2 represent the relative importance of different types of errors in the combined ASV and CM system. The default tandem Detection Cost Function is expressed as:

$$t\text{-DCF}_{\text{default}} = C_0 + \min(C_1, C_2) \quad (27)$$

These evaluation metrics—EER and $\min_t\text{-DCF}$ —quantify how well the system distinguishes between genuine and spoofed audio. In the context of spoofing detection, lower values of both metrics indicate better system performance, reflecting a more accurate and reliable classification of input audio as either original or fake.

4.3. Implementation details

The improved ResNeXt-based model was implemented using the PyTorch framework. The training procedure employed the Adam optimizer with a learning rate of 1×10^{-4} and a batch size of 32. The network was trained for 100 epochs, and the model checkpoint corresponding to the lowest validation EER was selected for evaluation. All experiments were conducted on a single NVIDIA RTX 3090 GPU.

To assess the model's performance on the ASVspoof 2019 Logical Access (LA) dataset, a total of 25,380 audio samples were used for training, consisting of 2,580 bona fide and 22,800 spoofed utterances. The proposed method was evaluated on both the evaluation (eval) and development (dev) subsets of the ASVspoof 2019 LA dataset. The eval set comprised a total of 71,237 samples, including 7,355 spoofed and 63,882 bona fide utterances. The dev set consisted of 24,844 samples, of which 22,296 were spoofed and 2,548 were bona fide.

This evaluation setup enables a thorough assessment of the model's performance under both known and unseen spoofing conditions.

In addition to the experiments conducted on training and testing with the LA dataset, the model has also been trained and tested on the PA dataset, which is entirely distinct and includes replay attacks. We trained the model with 48600 spoof and 5400 bonafide samples and tested on eval and dev sets. The eval set consist of 143,000 audio samples, while dev set consist of 29,700 samples.

Table 3
Results of different fusion techniques.

System	Features	EER (%)	min t-DCF
Individual	LFCC	1.98	0.04
Individual	MFCC	6.11	0.15
Early feature concatenation	LFCC and MFCC	3.23	0.07
Classification score level fusion	LFCC and MFCC	1.15	0.03
Feature embedding concatenation	LFCC and MFCC	0.72	0.02

4.4. Ablation studies

This section presents performance evaluations using various fusion techniques, organized into subsections. The experiments were conducted on the ASVspoof 2019 LA dataset.

Data fusion integrates multiple sources to provide more accurate and reliable information than any single source. Within this domain, Score-Level and Feature-Level Fusion represent two pivotal strategies for combining data. Understanding their distinctions, strengths, and appropriate use-cases is crucial for researchers and practitioners aiming to optimize system performance in various applications, including biometric recognition, sensor networks, and machine learning ensembles.

Score-level fusion, also referred to as decision-level fusion, is a technique in which the output scores – typically representing probabilities or confidence values – produced by multiple classifiers or decision-making units are aggregated to produce a final prediction. Each classifier operates independently, relying on its own feature set, and their outputs are combined to improve overall decision accuracy.

In contrast, feature-level fusion integrates raw or pre-processed features extracted from multiple sources prior to any classification stage. This approach merges individual feature vectors into a single, unified representation, which is then used to train a classifier. By consolidating diverse information at the feature stage, this method aims to improve the model's discriminative power.

In the context of neural networks, feature-level fusion can be applied at different stages of the architecture. At the input level, features are concatenated into a single vector and passed through the network. Alternatively, fusion can occur at an intermediate stage: each feature set is processed independently to generate high-level embeddings, which are then concatenated before the final classification layer.

These three fusion strategies exhibit distinct advantages and trade-offs, each suited to different application scenarios.

Early feature concatenation is the most computationally efficient approach, as a single neural network processes the entire feature set jointly. However, this simplicity may reduce the model's capacity to capture and learn the specific characteristics of each individual feature type, potentially limiting overall performance.

In contrast, score-level fusion uses multiple independent classifiers, each tailored to the specific attributes of its corresponding feature set. While this enhances adaptability and allows for feature-specific specialization, it also increases computational cost due to the parallel execution of multiple models. Moreover, relying solely on a single scalar score from each classifier can limit the fusion process in effectively capturing the richness of the underlying representations.

Embedding-level fusion offers a compromise between these two extremes. In this approach, separate networks extract high-level embeddings from each feature source, preserving their unique structure. These embeddings are then concatenated before classification, enabling the system to benefit from both detailed feature representations and improved decision accuracy, while maintaining moderate computational complexity.

To determine which of the fusion strategies is most effective in our setting, we first trained and evaluated the model using LFCC and MFCC features independently. Subsequently, we assessed the fusion of these two features using the three previously described strategies.

Table 4
Comparison of different combination of feature embedding concatenation.

Features	EER (%)	min t-DCF
LFCC	1.98	0.04
CQCC	2.78	0.06
MFCC	6.11	0.15
LFCC+MFCC	0.72	0.02
LFCC+CQCC	1.14	0.03
MFCC+CQCC	0.89	0.02
LFCC+MFCC+CQCC	0.61	0.017

The Equal Error Rate (EER) and minimum tandem Detection Cost Function (min t-DCF) scores obtained for each approach are summarized in Table 3.

Early feature-level fusion resulted in an EER of 3.23% and a min t-DCF of 0.07. Score-level fusion achieved better results with an EER of 1.15% and a min t-DCF of 0.03. Embedding-level fusion outperformed both methods, yielding the best results with an EER of 0.72% and a min t-DCF of 0.02.

Given the superior performance of embedding-level fusion, we extended our experiments by combining feature embeddings from different feature types—individually, in pairs, and all three together. The results of these experiments are presented in Table 4. As shown in the table, the configuration that concatenated all three feature embeddings produced the best overall performance. Based on this outcome, the full embedding concatenation strategy was adopted as the final system configuration.

To determine which approach performed best in our scenario, we began by training and evaluating the model using LFCC and MFCC features individually. Subsequently, we investigated the fusion of these two feature sets using the three previously described strategies. The results, reported in terms of Equal Error Rate (EER) and minimum tandem Detection Cost Function (min t-DCF), are summarized in Table 3.

The early feature-level fusion method resulted in an EER of 3.23% and a min t-DCF of 0.07. In contrast, score-level fusion yielded improved performance with an EER of 1.15% and a min t-DCF of 0.03. Among all methods, embedding-level fusion achieved the best results, with an EER of 0.72% and a min t-DCF of 0.02.

Motivated by the superior performance of embedding fusion, we extended our experiments by combining embeddings derived from different feature sets. This analysis included single-feature, pairwise, and triplet embedding combinations in various configurations. The results of these experiments are presented in Table 4. As shown, the concatenation of three embeddings produced the most effective results. Accordingly, this configuration was selected as the final system.

4.5. Evaluation results on ASVspoof 2019 LA dataset

This section presents a detailed performance evaluation of individual and fused systems, supported by visual demonstrations and analysis based on spoofing attack types.

The first set of experiments was conducted using the ASVspoof 2019 LA evaluation set, with a focus on attack-wise performance. The results include individual systems based on CQCC, LFCC, and MFCC features, as well as their binary fusion (LFCC + MFCC) and the final system (MLP-based score-level fusion of LFCC, MFCC, and CQCC). The Equal Error Rate (EER) for each spoofing attack type is reported in Table 5. For each attack, the lowest EER is highlighted.

In general, the binary fusion of LFCC and MFCC significantly improves performance across most attacks. For instance, in the case of attack A10, the EER drops dramatically from 16.610 (MFCC) to 0.750 with binary fusion. Similar patterns are observed for other attacks. The final system further enhances the detection performance by producing EER values that are either comparable to or better than binary fusion. Notably, it achieves an EER of 0.220 for attacks A08, A09, and A12.

Table 5
Comparison of individual and fused systems with EER (%).

Labels	CQCC	LFCC	MFCC	Binary concat.	Final system
A07	0.098	0.139	17.214	0.105	0.122
A08	0.122	0.098	0.587	0.017	0.122
A09	0.0407	0.139	0.139	0.488	0.081
A10	0.407	0.383	16.610	0.384	0.407
A11	0.285	0.227	2.583	0.081	0.081
A12	0.122	0.139	0.896	0.139	0.122
A13	0.139	0.179	0.716	0.122	0.122
A14	0.488	0.448	2.030	0.179	0.163
A15	0.505	0.587	4.149	0.179	0.163
A16	0.302	0.349	0.896	0.308	0.261
A17	15.119	9.951	6.610	2.648	2.339
A18	0.546	1.222	1.728	0.856	0.506
A19	1.704	0.937	2.933	0.612	0.465
Overall	2.787	1.984	6.11	0.723	0.626

Table 6
Comparison of EER (%) and min t-DCF results for different feature extraction methods.

Approach	TTS		VC		TTS-VC	
	EER	min t-DCF	EER	min t-DCF	EER	min t-DCF
CQCC	0.196	0.005	5.790	0.268	0.378	0.010
LFCC	0.196	0.005	4.037	0.209	0.405	0.011
MFCC	5.547	0.142	3.757	0.194	2.298	0.062
Binary concat.	0.134	0.003	1.372	0.090	0.160	0.004
Final system	0.136	0.003	1.103	0.067	0.149	0.003

Attack A17 remains the most challenging scenario across all systems, with relatively higher EERs. However, even in this case, fusion methods contribute to a performance gain. Overall, binary fusion reduces the average EER to 1.101, while the final system further improves it to 1.050.

4.5.1. Robustness against different attacks on ASVspoof 2019 LA dataset

The synthetic audio samples in the ASVspoof 2019 LA dataset are generated using a variety of spoofing systems. This section presents an analysis of system performance based on these spoofing methods.

Specifically, samples labeled A07–A12 and A16 are generated using text-to-speech (TTS) techniques, while A17 and A19 are produced using voice conversion (VC). Samples A13 and A15 are created using hybrid methods that combine both VC and TTS approaches.

The performance of the proposed system across these spoofing categories is summarized in Table 6, in terms of Equal Error Rate (EER) and minimum tandem Detection Cost Function (min t-DCF).

For TTS-based attacks, the system demonstrates strong robustness, achieving an EER of 0.196% and a min t-DCF of 0.005. These results suggest that features such as CQCC and LFCC are highly effective in detecting synthetic speech generated by TTS algorithms.

When evaluating overall system performance, the final fused model delivers the best generalization across all spoofing types. For TTS-based attacks, it achieves an EER of 0.283% and a min t-DCF of 0.008, while for VC-based attacks, it obtains an EER of 2.002% and a min t-DCF of 0.125. For hybrid spoofing systems, the model achieves an EER of 0.302% and a min t-DCF of 0.007, representing the lowest values observed across all evaluated methods.

These results indicate that the final system offers strong generalization capabilities, particularly in handling a diverse range of spoofing techniques.

The attack types included in the evaluation set of the ASVspoof 2019 LA dataset are categorized as known, unknown, or partially known based on their presence in the training and development subsets. Specifically, A16 and A19 are categorized as known attacks, A10–A15 and A18 as unknown attacks, and A07–A09 and A17 as partially known attacks.

The performance of individual systems, binary fusion, and the final fusion model across these three categories is summarized in Table 7.

Table 7
Comparison of EER (%) and min t-DCF for different approaches.

Approach	Known		Unknown		Partially Known	
	EER	min t-DCF	EER	min t-DCF	EER	min t-DCF
CQCC	1.00	0.033	0.35	0.011	3.84	0.181
LFCC	0.64	0.019	0.45	0.076	2.55	0.076
MFCC	1.91	0.058	4.10	0.114	6.11	0.213
Binary concat.	0.46	0.014	0.265	0.009	0.688	0.053
Final system	0.36	0.010	0.219	0.006	0.615	0.042

For known attacks, the final system achieves the best performance, with an Equal Error Rate (EER) of 0.36% and a minimum tandem Detection Cost Function (t-DCF) of 0.013. These results indicate higher accuracy and lower error cost when compared to individual feature-based systems.

For unknown attacks, the final system also outperforms all other approaches, achieving the lowest EER of 0.32% and a t-DCF of 0.010. These values highlight the model's ability to generalize effectively to previously unseen spoofing methods.

In the case of partially known attacks, which are generally more challenging due to limited prior information, the final system again demonstrates strong performance with an EER of 1.47% and a t-DCF of 0.089. Although the binary fusion approach yields a slightly lower EER of 1.46%, the final system still offers more balanced and reliable performance overall.

In summary, while the binary fusion approach also substantially improves performance over individual systems, the final system consistently achieves the best results across all attack categories. This makes it the most robust and dependable configuration for spoofing detection.

4.5.2. Comparison with state-of-the-art methods on ASVspoof 2019 LA

This section presents the evaluation results on the ASVspoof 2019 LA and PA datasets, comparing the proposed fusion-based strategy with existing methods in the literature. The goal is to assess the relative performance of the proposed system in a broader context.

The performance results on the LA dataset for both the proposed final system and several baseline and recent approaches are summarized in Table 8. The comparison is based on Equal Error Rate (EER) and minimum tandem Detection Cost Function (min t-DCF) metrics.

Baseline systems reported in the ASVspoof 2019 Challenge [4] exhibit relatively high EER values, ranging from 8.09% to 9.57%. The method proposed by Wang et al. (2020) [29], which combines CQCC features with a DenseNet architecture, performs worse, reporting an EER of 10.87%.

In contrast, more recent approaches have demonstrated improved performance. For example, the method by Xue et al. (2023) achieves an EER of 1.64%, while Wang et al. (2024) reports an even lower EER of 1.11%.

Despite these advancements, the proposed fusion-based approach outperforms all compared methods, delivering lower EER and min t-DCF values. These results demonstrate the effectiveness and reliability of the final fused system in detecting spoofed audio on the LA dataset.

4.6. Evaluation results on ASVspoof 2019 PA dataset

This section presents the evaluation results of individual features and fusion techniques on the ASVspoof 2019 PA dataset. The PA dataset is specifically designed to simulate replay attacks and was recorded using a stationary microphone in a controlled acoustic environment. The dataset comprises nine distinct replay attack types, labeled AA, AB, ..., CB, CC. These variations are defined based on two primary factors: the distance between the speaker (or playback source) and the microphone, and the quality of the recording and playback devices used during the spoofing process. These controlled conditions allow

Table 8
State-of-the-art methods and their performance on ASVspoof 2019 LA dataset.

State-of-the-art methods	Main approach basically	EER (%)	min t-DCF
Todisco et. al. (2019) [4]	LFCC-GMM	8.09	0.21
Todisco et. al. (2019) [4]	CQCC-GMM	9.57	0.23
Alzantot et. al. (2019) [15]	Log-STFT-Resnet	3.81	0.10
Alzantot et. al. (2019) [15]	MFCC-Resnet	9.33	0.20
Alzantot et. al. (2019) [15]	CQCC-Resnet	7.69	0.22
Wu et. al. (2020) [28]	FG+LCNN	4.07	0.10
Wang et. al. (2020) [29]	MFCC-DenseNet	9.64	0.22
Wang et. al. (2020) [29]	CQCC-DenseNet	10.87	0.26
Wang et. al. (2020) [29]	Spec-DenseNet	7.14	0.20
Zhang et. al. (2021) [19]	LFCC+Resnet18	2.19	0.05
Li et. al. (2021) [27]	CQT+SCG-Res2Net50	2.43	0.07
Li et. al. (2021) [27]	CQT & MCG-Res2Net50	1.78	0.05
Li et. al. (2021) [27]	CQT & MLCG-Res2Net50	2.15	0.06
Hak et. al. (2021) [55]	RawNet2	4.62	0.12
Hua et. al. (2021) [56]	Res-TSSDNet	1.64	0.04
Jung et. al. (2022) [54]	AASIST	0.83	0.02
Jung et. al. (2022) [54]	AASIST-L	0.99	0.03
Xue et. al. (2023) [30]	LFCC&Face+SE-DenseNet+A-softmax	2.73	0.07
Xue et. al. (2023) [31]	LFCC&Face+SE-DenseNet+CE	2.82	0.07
Grinberg et. al. (2023) [6]	RECAPA-TDNN	1.11	0.07
Xue et. al. (2023) [31]	LFCC&Face+SE-DenseNet+CE	2.82	0.07
Zaman et. al. (2024) [7]	VGG-Transformer	8.29	–
Zaman et. al. (2024) [7]	ResNet-Transformer(Adam)	7.97	–
Zaman et. al. (2024) [7]	ResNet-Transformer(RMSProp)	3.22	–
Wang et. al. (2024) [32]	LFCC+MPE - LCNN	2.41	–
Wang et. al. (2024) [32]	LFCC+MPE - SENet	1.94	–
Chaiwongyen et al. (2024) [33]	PF+Melspec+ResNet18	15.97	–
Mirza et. al. (2024) [8]	Opensmile - SVM	7.71	0.22
Mirza et. al. (2024)[8]	Opensmile - DNN	6.67	0.21
Proposed method	Embedding concatenation	0.61	0.017

Table 9
Comparison of EER(%) and min t-DCF for different feature combinations on PA dataset.

	MFCC		Binary concat.		Final system	
	EER	min t-DCF	EER	min t-DCF	EER	min t-DCF
AA	2.50	0.06	1.701	0.047	1.753	0.048
AB	0.97	0.02	0.648	0.018	0.631	0.018
AC	0.82	0.02	0.514	0.014	0.573	0.014
BA	2.13	0.05	1.329	0.036	1.388	0.037
BB	0.95	0.02	0.591	0.017	0.616	0.016
BC	0.68	0.02	0.419	0.013	0.480	0.014
CA	2.18	0.06	1.382	0.036	1.398	0.037
CB	1.02	0.02	0.593	0.016	0.608	0.0159
CC	0.81	0.02	0.470	0.013	0.529	0.0152
Overall	1.5	0.04	1.00	0.029	0.99	0.028

for systematic analysis of the model's ability to detect replay-based spoofing under different acoustic and hardware configurations.

The evaluation results for each replay attack type in the ASVspoof 2019 PA dataset are presented in Table 9. In the MFCC-based system, the highest Equal Error Rate (EER) is observed for attack type AA at 2.50%, while the lowest is recorded for attack type AC at 0.82%. The binary fusion approach consistently reduces EER values across attack types. For example, attack type BB achieves the lowest EER of 0.75%, while the highest remains with type AA at 2.02%. Further improvements are observed with the final system, which demonstrates reduced EERs for all attack types. Notably, the EER for attack type AA decreases to 1.95%, while the lowest EER of 0.82% is achieved for both BB and BC. Pooled results across all replay conditions confirm the effectiveness of the proposed approach. The final system achieves the lowest overall EER of 1.14% and a minimum tandem Detection Cost Function (min t-DCF) of 0.03, indicating superior robustness in detecting replay-based spoofing attacks under diverse acoustic and device configurations.

Table 10 provides a comparison of various state-of-the-art methods evaluated on the ASVspoof 2019 PA dataset, with performance measured in terms of EER and min t-DCF. Some studies that reported

results on the LA dataset did not provide results for the PA dataset; therefore, they were excluded from this comparison. Recent methods, particularly those utilizing deep learning frameworks such as ResNet and DenseNet, demonstrated superior performance, with Alzantot et al. (2019) [15] achieving a lower EER of 4.43%. The proposed fused approach achieved superior and satisfactory results, with an EER of 1.14% and a min t-DCF of 0.33, on the ASVspoof 2019 PA dataset.

4.7. Evaluation results on ASVspoof 2021 dataset

This section presents the test results on the ASVspoof 2021 LA, PA and DF datasets. Fig. 4 provides an analysis of conditions and attacks, with attack algorithms represented on the vertical axis and evaluation conditions on the horizontal axis on ASVspoof 2021 LA set. The color of each circle indicates the median minimum t-DCF for the top-10 systems, while the radius of the circle denotes the interquartile range. Certain attacks and conditions consistently exhibit high minimum t-DCF values. For instance, attack A17, and to a lesser extent A18, show significant variability across different conditions. Specifically, the median t-DCF for A17 ranges from approximately 0.8 under condition C1 to over 0.9 under conditions C2, C3, and C7. These results emphasize that attack A17, which is already challenging to detect even without encoding, becomes substantially more difficult to identify after the application of low-bandwidth encoding.

The proposed framework is compared against four state-of-the-art (SOTA) methods: CQCC-GMM [5], LFCC-GMM [5], LFCC-LCNN [9], and RawNet2 [55]. The comparison is conducted across individual evaluation conditions (C1 to C7), as well as pooled across all attack scenarios.

Tables 11 and 12 present the SOTA results reported in [69] alongside those obtained with the proposed method for reference on ASV Spoof 2021 LA set.

Table 11 shows that our approach achieves superior performance compared to these baselines, obtaining the lowest EER for all attack types across every condition, with a pooled EER of 7.44%. This significantly outperforms baseline systems such as CQCC-GMM (15.62%),

Table 10
State-of-the-art methods and their performance on ASVspoof 2019 PA dataset.

State-of-the-art methods	Main approach basically	EER (%)	min t-DCF
Todisco et. al. (2019) [4]	LFCC-GMM	13.54	0.30
Todisco et. al. (2019) [4]	CQCC-GMM	11.04	0.24
Alzantot et. al. (2019) [15]	Log-STFT-Resnet	9.68	0.27
Alzantot et. al. (2019) [15]	CQCC-Resnet	4.43	0.10
Wang et. al. (2020) [29]	CQCC-DenseNet	6.40	0.15
Wang et. al. (2020) [29]	Spec-DenseNet	3.01	0.08
Wang et. al. (2021) [9]	LFCC & LCNN	9.73	0.31
Wang et. al. (2021) [9]	RawNet2	8.05	0.30
Wang et. al. (2021) [9]	Fusion (LFCC & LCNN and RawNet2)	7.88	0.30
Xue et. al. (2023) [30]	LFCC - Face+SE-DenseNet+A-softmax	2.73	0.07
Mirza et. al. (2024) [8]	Opensmile - SVM	7.71	0.22
Mirza et. al. (2024) [8]	Opensmile - DNN	6.67	0.21
Proposed method	Embedding concatenation	0.99	0.028

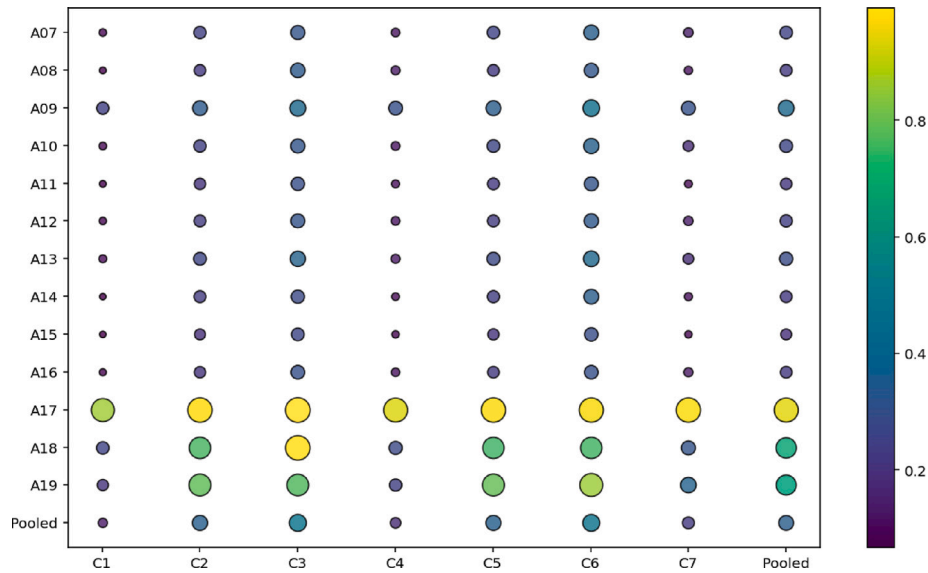


Fig. 4. Statistics on min-tDCF values of top-10 ASVspoof 2021 LA submissions in sub-conditions. The size of each dot corresponds to the interquartile range of the minimum t-DCF distribution, while its color represents the median value of the distribution.

LFCC-GMM (19.30%), LFCC-LCNN (9.26%), and RawNet2 (9.50%). Notably, our method demonstrates consistent improvements, particularly in challenging conditions such as C3 and C5, where the EER is reduced to 11.93% and 7.69%, respectively, compared to substantially higher rates observed in baseline systems. These results highlight the robustness and effectiveness of the proposed approach in addressing spoofing detection challenges across diverse conditions.

Table 12 compares our framework's performance against baseline systems using the minimum t-DCF metric across individual conditions (C1 to C7) as well as pooled over all attacks on this dataset. Our method achieves a pooled min t-DCF of 0.35, outperforming referenced baseline approaches including CQCC-GMM (0.49), LFCC-GMM (0.57), LFCC-LCNN (0.34), and RawNet2 (0.42). Additionally, it consistently performs well across most conditions, exhibiting notable improvements in C1 (0.13) and C4 (0.18). Although slightly higher min t-DCF values are observed under conditions C2 (0.37) and C6 (0.46) compared to LFCC-LCNN, the overall results indicate that the proposed methodology attains the best performance in the majority of cases, underscoring its effectiveness for spoofing detection.

We achieved improved overall performance on the DF eval set, the EER(%) results are given in Table 13. For individual DF attacks, marginal improvements over the baseline system were observed on C1, C3 and C8.

From the last experiment, Table 14 showcases the performance comparison of various systems on the ASVspoof 2021 PA dataset. The evaluated systems include traditional and state-of-the-art models such

Table 11

EER (%) results of the methods over attacks on ASVspoof 2021 LA. SOTA results are gathered from [69] as reference.

Methods	C1	C2	C3	C4	C5	C6	C7	Pooled
CQCC-GMM [5]	10.57	14.76	20.58	11.61	13.58	14.01	11.21	15.62
LFCC-GMM [5]	12.72	21.21	35.55	15.28	18.76	18.46	12.73	19.30
LFCC-LCNN [9]	6.71	8.89	12.02	6.34	9.25	11.00	6.66	9.26
RawNet2 [55]	5.84	6.59	16.72	6.41	6.33	10.66	7.95	9.50
Proposed method	2.61	8.11	11.93	3.82	7.69	10.14	5.82	7.44

Table 12

Min t-DCF results of the methods over attacks on ASVspoof 2021 LA. SOTA results are gathered from [69] as reference.

Methods	C1	C2	C3	C4	C5	C6	C7	Pooled
CQCC-GMM [5]	0.28	0.51	0.61	0.37	0.47	0.47	0.29	0.49
LFCC-GMM [5]	0.36	0.63	0.83	0.44	0.58	0.60	0.36	0.57
LFCC-LCNN [9]	0.21	0.33	0.42	0.22	0.35	0.42	0.21	0.34
RawNet2 [55]	0.20	0.35	0.63	0.23	0.34	0.49	0.29	0.42
Proposed method	0.13	0.37	0.47	0.18	0.37	0.46	0.22	0.35

as CQCC-GMM, LFCC-GMM, LFCC-LCNN, and RawNet2. Among these, the proposed method achieves the lowest EER of 32.35%, significantly outperforming all baseline systems in terms of spoofing detection accuracy. While the min t-DCF value of the proposed method (0.9766) is slightly higher than some of the baseline methods, its notable reduction

Table 13

EER (%) results of the methods over attacks on ASVspoof 2021 DF. SOTA results are gathered from [69] as reference.

System	C1	C2	C3	C4	C5	C6	C7	C8	C9	Pooled
CQCC-GMM [5]	19.48	48.86	20.37	19.55	20.27	17.92	14.42	49.41	17.39	25.56
LFCC-GMM [5]	17.39	39.20	17.97	20.95	21.43	22.16	14.83	39.03	26.65	25.25
LFCC-LCNN [9]	23.19	34.21	23.88	25.22	23.85	19.06	17.10	28.35	18.54	23.48
RawNet2[55]	26.98	27.63	27.49	26.72	27.23	18.80	18.67	18.74	19.10	22.38
Proposed method	12.64	26.82	13.02	21.09	20.44	15.32	14.24	14.58	21.12	18.43

Table 14

Min t-DCF and EER (%) results of the methods on ASVspoof 2021 PA. SOTA results are gathered from [70] as reference.

System	Min t-DCF	EER (%)
CQCC-GMM [5]	0.9434	38.07
LFCC-GMM [5]	0.9724	39.54
LFCC-LCNN [9]	0.9958	44.77
RawNet2 [55]	0.9997	48.60
Proposed	0.9766	32.35

in EER highlights its practical advantage in real-world scenarios, where minimizing false positives and false negatives is critical. This result demonstrates the robustness and effectiveness of the proposed approach against physical access attacks in speaker verification systems.

5. Discussion

This study introduces a deepfake audio detection system employing a ResNeXt-based architecture that leverages embedding-level fusion of LFCC, MFCC, and CQCC spectral features. Experimental results obtained from three challenging ASVspoof datasets (2019 LA, 2019 PA, and 2021 LA, PA, and DF) demonstrate that the proposed embedding-level fusion approach significantly enhances detection performance compared to systems using individual features or early fusion strategies.

A deeper analysis on attack-type-level performance revealed that the effectiveness of spectral features varies depending on the type of spoofing technique employed:

- On the ASVspoof 2019 LA dataset, where TTS-based attacks were evaluated, the worst performance among individual features was obtained using MFCCs, whereas LFCC and CQCC showed comparable and much better results. Performance improved further when using binary feature fusion, and the best results were achieved with the final system that combines all three features.
- In contrast, for VC-based attacks, the worst individual result was obtained using CQCC, while surprisingly, MFCC alone yielded the best individual performance. However, even in this case, the final fused system outperformed all individual and binary combinations, demonstrating the strength of embedding-level feature fusion.
- On the ASVspoof 2019 PA dataset, which contains replay attacks, the best individual performance was achieved using MFCC features, indicating that the model benefits from the temporal dynamics captured by MFCCs in these types of attacks.

These findings underline the complementary nature of different spectral features and show that the proposed final fusion system is capable of robust performance across diverse spoofing methods. This confirms the generalization ability of the proposed model — it can accurately detect fake audio regardless of whether it was generated using text-to-speech (TTS), voice conversion (VC), or replay attacks. While the system reflects high accuracy, especially in detecting TTS and VC-based attacks, there is one important factor that needs to be given more importance and that is model interpretability. It is crucial to understand why a model classifies a specific audio as spoofed or bonafide since this is extremely essential for trustworthy deployment, especially in high-risk applications like biometric authentication and forensic analysis.

Recently, several studies have reported notable improvements in deepfake audio detection by leveraging pre-trained SSL models such as HuBERT [57], Wav2Vec2 [58], and WavLM [59], or by augmenting training data using vocoders [60]. While these approaches have shown promising results, it is important to note that most of these works do not report performance on critical datasets such as ASVspoof 2019 PA or 2021 PA, which contain replay attacks—a particularly challenging scenario in real-world applications. For instance, the study utilizing multiple SSL-based embeddings (HuBERT-Base, Wav2Vec2-Large, and WavLM-Base) [60] avoids these datasets, raising questions about the generalization capacity of these methods in replay attack detection. In contrast, our proposed system achieves competitive performance across all three types of spoofing attacks (TTS, VC, and replay) without relying on pre-trained models or vocoder-generated data. This reinforces the strength and generalizability of our embedding-level fusion approach and highlights its practical relevance, especially in contexts where labeled data is scarce or where black-box dependencies are undesirable. To further understand the system's behavior in real-world scenarios, we also evaluated its inference behavior using a test audio sample with a duration of approximately **1.33 minutes**. The system was able to process this input efficiently without noticeable latency, suggesting its potential applicability in real-time or near-real-time deepfake audio detection pipelines. To assess the system's real-time inference behavior, we conducted a practical evaluation using 10 randomly selected utterances from the ASVspoof 2019 LA eval set. The test was performed on a laptop with an Intel i7 processor, 64 GB of RAM, and a GeForce RTX 3060 GPU. The average duration of these audio samples was approximately **1.33 minutes**. The system processed these inputs efficiently and without noticeable latency, indicating its potential viability for deployment in real-time or large-scale detection pipelines.

Though deep neural networks are black-box models by nature, recent breakthroughs in explainable AI (XAI) open up opportunities for model transparency improvement. In our case, spectral features such as CQCC are known to capture minor discrepancies introduced by synthesis methods. We therefore assume that the model learns to attend to unnatural spectral-temporal structures common in deepfake audio.

While the current system demonstrates high performance across a wide range of spoofing attacks, future work will focus on enhancing the model's robustness, under more challenging scenarios. Deepfake audio detection systems remain susceptible to adversarial examples—carefully perturbed audio inputs designed to mislead the model while remaining perceptually indistinguishable to human listeners. Future research will explore adversarial robustness through techniques such as adversarial training, where the model is exposed to perturbed inputs during learning, input randomization, and certified defenses that provide theoretical guarantees of resistance against small adversarial manipulations. These strategies will ensure reliable operation in security-critical environments.

6. Conclusions

In this study, a comprehensive approach to audio spoof detection is proposed, utilizing LFCC, MFCC, and CQCC spectral features alongside an improved ResNeXt-based architecture to obtain feature embeddings. The embeddings from each feature type are concatenated and classified using an MLP-based approach. Our method outperforms existing

methods on the ASVspoof 2019 LA, ASVspoof 2019 PA, and ASVspoof 2021 LA datasets, achieving superior performance metrics, including an EER of 1.05% and a min t-DCF of 0.028 for ASVspoof 2019 LA, and an EER of 1.14% and a min t-DCF of 0.033 for ASVspoof 2019 PA. The MLP-based classification of concatenated feature embeddings proved effective in distinguishing between genuine and spoofed audio, even across various attack types such as text-to-speech and voice conversion. These results highlight the robustness and versatility of the proposed system in addressing evolving challenges in deepfake audio detection. Future work could explore real-time detection capabilities and evaluate the model's effectiveness on additional datasets and diverse attack scenarios.

CRedit authorship contribution statement

Gul Tahaoglu: Writing – review & editing, Writing – original draft, Software, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Daniele Baracchi:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Dasara Shullani:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Conceptualization. **Massimo Iuliani:** Writing – review & editing, Conceptualization. **Alessandro Piva:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Gul Tahaoglu reports financial support was provided by TUBITAK. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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